

Supplementary Methods: Quantifying Facial Morphological Idiosyncrasy via Manifold-Regularised Structural Connectomics

This supplement sets out, in full, the mathematical pipeline used to quantify facial structural idiosyncrasy at two nested levels — the individual triangular facet and the whole face — within a cohort of Northwestern Han Chinese participants. Each participant’s face is represented as a dense, deep-learning-derived structural connectome defined over the facets of the hemi-face, and idiosyncrasy is measured as the statistical divergence of a participant’s connectivity pattern from the cohort baseline.

Because the number of facets greatly exceeds the cohort size, the empirical facet covariance is rank-deficient and cannot be inverted directly, which precludes a naïve Mahalanobis treatment. The pipeline resolves this by replacing linear covariance shrinkage with a non-linear, data-driven regularisation scheme: the baseline covariance is confined to an intrinsic geometric manifold recovered by diffusion-map embedding, and an isotropic residual term is added to guarantee strict positive-definiteness and hence global invertibility.

The document is organised as a self-contained development. Sections 2–4 construct the ingredients of the model; Section 5 defines the manifold-regularised covariance operator and establishes its validity through two lemmata with complete proofs; Sections 6–7 define the resulting facet-level and individual-level idiosyncrasy scores; and Section 8 collects the well-posedness guarantees carried into the downstream connectome analysis.

1. Problem Setting and Notational Conventions

We study a cohort of N individuals (the cohort size, a scalar), indexed by $i = 1, \dots, N$. The hemi-face of each individual is discretised into a fixed triangular mesh of $F = 7,025$ facets (a scalar), indexed by $f, j, m \in \{1, \dots, F\}$. The defining structural characteristic of the problem is the strong inequality

$$F = 7,025 > N,$$

which renders any per-facet empirical covariance matrix singular and motivates the regularisation developed below.

Throughout, lower-case bold symbols (for example $\mathbf{c}$, $\boldsymbol{\mu}$) denote real column vectors, and upper-case symbols (for example C , Σ) denote real matrices. The notation $\mathbb{R}^d$ denotes the space of d -dimensional real column vectors, and $\mathbb{R}^{p \times q}$ the space of $p \times q$ real matrices. We write I_d for the $d \times d$ identity matrix, $(\cdot)^T$ for transposition, $\|\cdot\|$ for the Euclidean (ℓ_2)

norm, $\text{Tr}(\cdot)$ for the matrix trace, $\text{rank}(\cdot)$ for the rank, $\det(\cdot)$ for the determinant, and $A \succcurlyeq 0$ to indicate that a symmetric matrix A is positive semi-definite.

2. Individual-Level Extraction of the Facet Connectivity Profile

For each individual i , the deep-learning pipeline yields a structural connectome matrix

$$C_i \in \mathbb{R}^{F \times F},$$

a symmetric matrix whose entry $C_i(j, f)$ encodes the (possibly non-linear) structural interaction between facets j and f of that individual's face. The full matrix therefore captures the complete pairwise wiring diagram over all $F = 7,025$ facets.

To isolate the global connectivity signature of a single target facet f , we extract its corresponding column from C_i , giving the **facet connectivity profile vector**

$$\mathbf{c}_{i,f} = \begin{bmatrix} C_i(1, f) \\ C_i(2, f) \\ \vdots \\ C_i(F, f) \end{bmatrix} \in \mathbb{R}^F.$$

Each $\mathbf{c}_{i,f}$ is an F -dimensional vector: it records how facet f of individual i is wired to every facet of that individual's face, and it is the elementary observation on which all subsequent statistics are built.

3. Group-Level Baseline Distribution of the Connectivity Profile

Using the entire cohort of N individuals, we characterise the background distribution of connectivity for each facet f through its first two moments.

Population mean profile. The **group-level mean connectivity profile vector** for facet f is

$$\boldsymbol{\mu}_f = \frac{1}{N} \sum_{i=1}^N \mathbf{c}_{i,f} \in \mathbb{R}^F,$$

an F -dimensional vector giving the cohort-average wiring pattern of facet f .

Population covariance. The **empirical population covariance matrix** for facet f is

$$\Sigma_f = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)(\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)^T \in \mathbb{R}^{F \times F},$$

a symmetric, positive semi-definite matrix describing how the connectivity of facet f covaries across the cohort. Because it is assembled from N outer products of F -dimensional vectors with $F > N$, the matrix Σ_f has rank at most $N-1$ and is therefore singular: it cannot be inverted, and the Mahalanobis distance is undefined without regularisation. Overcoming this deficiency is the purpose of Sections 4 and 5.

4. Non-Linear Manifold Discovery via Diffusion-Map Embedding

Facial connectivity is constrained by non-linear anatomical boundaries, so the population covariance for a facet is expected to concentrate near a low-dimensional, curved subset of $\mathbb{R}^F$ rather than to fill the space linearly. We recover this intrinsic subset — the **facial connectivity manifold** — by a diffusion-map embedding computed over the F connectivity channels, treating the rows of Σ_f as channel feature descriptors. The procedure comprises four stages.

4.1 Affinity matrix. We first define a local similarity between connectivity channels j and m using a Gaussian kernel applied to the distance between their covariance profiles (that is, the corresponding rows $\Sigma_f(j, \cdot)$ and $\Sigma_f(m, \cdot)$ of Σ_f):

$$A_f(j, m) = \exp\left(-\frac{\|\Sigma_f(j, \cdot) - \Sigma_f(m, \cdot)\|^2}{\varepsilon}\right), \quad A_f \in \mathbb{R}^{F \times F}.$$

Here $\varepsilon > 0$ is a scalar scale hyperparameter, set to the median of the pairwise squared distances so that the kernel adapts to the empirical spread of the channel descriptors.

4.2 Density normalisation. To decouple the recovered manifold geometry from non-uniform sampling density across the face, we normalise the kernel by its degree. Let

$$D_{jj} = \sum_{m=1}^F A_f(j, m), \quad D = \text{diag}(D_{11}, \dots, D_{FF}) \in \mathbb{R}^{F \times F},$$

be the diagonal degree matrix of A_f . Adopting the density-normalisation parameter $\alpha = 1$ yields the density-corrected kernel

$$\tilde{A}_f = D^{-1} A_f D^{-1} \in \mathbb{R}^{F \times F}.$$

4.3 Markov transition matrix. We then form the row-stochastic transition matrix of the associated diffusion process. Let $\tilde{D}$ be the diagonal degree matrix of $\tilde{A}_f$, with entries $\tilde{D}_{jj} = \sum_{m=1}^F \tilde{A}_f(j, m)$. The Markov transition matrix is

$$M_f = \tilde{D}^{-1} \tilde{A}_f \in \mathbb{R}^{F \times F},$$

whose rows sum to unity and whose entries define the one-step transition probabilities of a random walk over the F connectivity channels.

4.4 Spectral embedding and gradient selection. We compute the eigendecomposition of M_f . Its eigenvalues are denoted $\gamma_0 \geq \gamma_1 \geq \gamma_2 \geq \dots$; the leading pair is the trivial stationary mode $\gamma_0 = 1$ with a constant eigenvector, which carries no geometric information and is discarded. Retaining the top k non-trivial right eigenvectors — the **diffusion gradients** $\psi_1, \dots, \psi_k$ associated with $\gamma_1 \geq \dots \geq \gamma_k$ — we assemble the embedding matrix

$$\Psi_k = [\psi_1, \quad \psi_2, \quad \dots, \quad \psi_k] \in \mathbb{R}^{F \times k}, \quad k \ll F,$$

where k (a scalar) is the number of retained gradients. To ensure strict geometric orthogonality in the projection of Section 5, the columns of Ψ_k are orthonormalised so that

$$\Psi_k^T \Psi_k = I_k,$$

with I_k the $k \times k$ identity matrix. The column space of Ψ_k is the k -dimensional **manifold subspace** onto which the baseline covariance will be confined.

Notational caution. The symbols γ_ℓ above denote eigenvalues of the diffusion transition matrix M_f and are used only to rank the gradients for selection. They are distinct from the eigenvalues λ_j of the covariance matrix Σ_f introduced in Section 5, which quantify population variance and appear explicitly in the proofs.

5. The Manifold-Regularised Covariance Operator

We now construct an invertible surrogate for the singular empirical covariance Σ_f . The construction confines the admissible population variance to the manifold subspace recovered in Section 4 and populates the complementary null space with an isotropic residual floor. We refer to the result as the **manifold-regularised covariance operator**, $\Sigma_{f, \text{reg}}$: it regularises a singular empirical matrix by compressing its valid covariance structure onto a data-driven, non-linear manifold while filling the null space with a residual noise floor that secures invertibility.

5.1 Spectral Prerequisites

Since Σ_f is real, symmetric and positive semi-definite, the Spectral Theorem guarantees an exact eigendecomposition

$$\Sigma_f = V \Lambda V^T = \sum_{j=1}^F \lambda_j \mathbf{v}_j \mathbf{v}_j^T,$$

where:

- $\mathbf{v}_j \in \mathbb{R}^F$ is the j -th orthonormal eigenvector, defining a linear axis of population variance in the connectivity space;
- $\lambda_j \geq 0$ is the corresponding eigenvalue, quantifying the population variance projected along $\mathbf{v}_j$, ordered $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_F \geq 0$;
- $V = [\mathbf{v}_1, \dots, \mathbf{v}_F] \in \mathbb{R}^{F \times F}$ is the orthogonal matrix of eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_F) \in \mathbb{R}^{F \times F}$ the diagonal matrix of eigenvalues.

This decomposition is the analytic device through which the projection of Section 5.3 is made explicit.

5.2 Definition of the Operator

The manifold-regularised covariance operator is defined as

$$\boxed{\Sigma_{f, \text{reg}} = \Psi_k (\Psi_k^T \Sigma_f \Psi_k) \Psi_k^T + \sigma^2 I_F} \in \mathbb{R}^{F \times F}.$$

The construction has two additive terms with complementary roles:

- **The manifold-projection term** $\Psi_k (\Psi_k^T \Sigma_f \Psi_k) \Psi_k^T$. The inner factor $\Psi_k^T \Sigma_f \Psi_k \in \mathbb{R}^{k \times k}$ is the **manifold core**: it compresses the full $F \times F$ covariance into a compact $k \times k$ representation that tracks variance only along the diffusion gradients. Pre- and post-multiplication by Ψ_k and Ψ_k^T re-expands this clean signal back to the ambient $F \times$

F facet dimensions. Section 5.3 (Lemma 1) proves that this term is an orthogonal projection of Σ_f onto the manifold subspace.

- **The isotropic residual term** $\sigma^2 I_F$, where I_F is the $F \times F$ identity matrix and $\sigma^2 > 0$ is a scalar variance floor. Section 5.4 (Lemma 2) proves that this term restores strict positive-definiteness and hence global invertibility.

The scalar σ^2 captures the residual population variance discarded when the covariance is truncated to the k -dimensional manifold, and is defined as

$$\sigma^2 = \frac{\text{Tr}(\Sigma_f) - \text{Tr}(\Psi_k^T \Sigma_f \Psi_k)}{F - k}.$$

Its non-negativity, positivity in the relevant regime, and interpretation as an averaged residual variance are established next.

Proposition (Residual Variance Floor). The scalar σ^2 equals the mean population variance residing in the orthogonal complement of the manifold subspace. It satisfies $\sigma^2 \geq 0$, with $\sigma^2 > 0$ whenever the range of Σ_f is not entirely contained within that subspace.

Proof. Write $P_k = \Psi_k \Psi_k^T$ for the manifold projector (shown to be a symmetric idempotent in Lemma 1 below). By the cyclic property of the trace and idempotency of P_k ,

$$\text{Tr}(\Psi_k^T \Sigma_f \Psi_k) = \text{Tr}(\Sigma_f \Psi_k \Psi_k^T) = \text{Tr}(\Sigma_f P_k) = \text{Tr}(P_k \Sigma_f P_k).$$

Introducing the complementary orthogonal projector $I_F - P_k$ and expanding,

$$(F - k)\sigma^2 = \text{Tr}(\Sigma_f) - \text{Tr}(\Sigma_f P_k) = \text{Tr}((I_F - P_k)\Sigma_f(I_F - P_k)).$$

The matrix $(I_F - P_k)\Sigma_f(I_F - P_k)$ is a congruence transform of the positive semi-definite matrix Σ_f and is therefore itself positive semi-definite, so its trace is non-negative; hence $\sigma^2 \geq 0$. The trace vanishes only if $(I_F - P_k)\Sigma_f(I_F - P_k) = 0$, that is, only if the entire range of Σ_f lies within the manifold subspace. Otherwise $\sigma^2 > 0$. Finally $F - k > 0$ because $k \ll F$, so σ^2 is well defined. In the empirical regime considered here, where k is a strict truncation of a covariance with support outside the manifold, σ^2 is strictly positive. ■

The identity above also lends σ^2 its interpretation: it is precisely the average eigen-energy of Σ_f discarded by the truncation, distributed uniformly across the $F - k$ complementary directions — a construction in the spirit of the noise-variance estimate of probabilistic principal component analysis.

5.3 Lemma 1 — The Projection Term is an Orthogonal Manifold Projection

Lemma 1 (Orthogonal Subspace Projection). The linear operator $\mathcal{P}_k(\Sigma_f) = \Psi_k(\Psi_k^T \Sigma_f \Psi_k)\Psi_k^T$ is the orthogonal projection of the full-dimensional empirical covariance Σ_f onto the k -dimensional diffusion manifold subspace. Equivalently,

it retains each population axis of variation only through its component expressible within that subspace.

Proof. Define the matrix

$$P_k = \Psi_k \Psi_k^T \in \mathbb{R}^{F \times F}.$$

We first show P_k is an orthogonal projector, that is, symmetric and idempotent.

Symmetry.

$$P_k^T = (\Psi_k \Psi_k^T)^T = \Psi_k \Psi_k^T = P_k.$$

Idempotency. Using the column-orthonormality $\Psi_k^T \Psi_k = I_k$,

$$P_k^2 = (\Psi_k \Psi_k^T)(\Psi_k \Psi_k^T) = \Psi_k (\Psi_k^T \Psi_k) \Psi_k^T = \Psi_k I_k \Psi_k^T = \Psi_k \Psi_k^T = P_k.$$

Hence P_k is the orthogonal projector onto the column space of Ψ_k — the manifold subspace. The projection term therefore rewrites as a symmetric two-sided projection of Σ_f :

$$\Psi_k (\Psi_k^T \Sigma_f \Psi_k) \Psi_k^T = (\Psi_k \Psi_k^T) \Sigma_f (\Psi_k \Psi_k^T) = P_k \Sigma_f P_k.$$

Substituting the spectral decomposition $\Sigma_f = \sum_{j=1}^F \lambda_j \mathbf{v}_j \mathbf{v}_j^T$ and defining the projected eigenvector

$$\hat{\mathbf{v}}_j = P_k \mathbf{v}_j = \Psi_k \Psi_k^T \mathbf{v}_j \in \mathbb{R}^F$$

as the image of the j -th population axis within the manifold subspace, linearity of P_k gives

$$P_k \Sigma_f P_k = \sum_{j=1}^F \lambda_j (P_k \mathbf{v}_j)(P_k \mathbf{v}_j)^T = \sum_{j=1}^F \lambda_j \hat{\mathbf{v}}_j \hat{\mathbf{v}}_j^T.$$

Each population axis $\mathbf{v}_j$ is thus retained only through its manifold-aligned component $\hat{\mathbf{v}}_j$, with its variance λ_j preserved along that constrained direction. This is the asserted orthogonal manifold projection. ■

Interpretation. The projected vector $\hat{\mathbf{v}}_j$ is the anatomical shadow of the j -th population axis of variation. A pattern of variation arising from landmark-digitisation noise or non-anatomical artefact tends to lie perpendicular to the diffusion gradients; its projection contracts towards zero and the pattern is suppressed. A pattern that follows authentic craniomaxillofacial growth or muscular structure aligns with the gradients; its variance λ_j is preserved cleanly, but is now constrained to act along the curved geometry of the face rather than in arbitrary linear directions.

5.4 Lemma 2 — Isotropic Perturbation Restores Invertibility

Lemma 2 (Resolution of Singularity via Isotropic Perturbation). If $k < F$, the projected matrix $\Sigma_{\text{filter}} = P_k \Sigma_f P_k$ is singular, with rank at most k . Adding the isotropic term $\sigma^2 I_F$ with $\sigma^2 > 0$ lifts the null-space eigenvalues uniformly, producing a strictly positive-definite operator $\Sigma_{f, \text{reg}}$ that is non-singular and globally invertible.

Proof. Write $\Sigma_{\text{filter}} = P_k \Sigma_f P_k$.

Rank and singularity. Because $\Psi_k \in \mathbb{R}^{F \times k}$ has $\text{rank}(\Psi_k) = k$, and the rank of a matrix product cannot exceed the rank of any factor,

$$\text{rank}(\Sigma_{\text{filter}}) = \text{rank}(\Psi_k [\Psi_k^T \Sigma_f \Psi_k] \Psi_k^T) \leq \min(\text{rank}(\Psi_k), \text{rank}(\Sigma_f)) \leq k.$$

Since $k < F$, the null space $\mathcal{N}(\Sigma_{\text{filter}}) = \{\mathbf{x} \in \mathbb{R}^F : \Sigma_{\text{filter}} \mathbf{x} = \mathbf{0}\}$ has dimension at least $F - k \geq 1$. For any non-zero $\mathbf{x} \in \mathcal{N}(\Sigma_{\text{filter}})$ the quadratic form collapses, $\mathbf{x}^T \Sigma_{\text{filter}} \mathbf{x} = 0$; equivalently, Σ_{filter} carries at least $F - k$ eigenvalues exactly equal to zero and is therefore singular and non-invertible.

Positive semi-definiteness of the filter. Since $\Sigma_f \succcurlyeq 0$ it admits a symmetric square root $\Sigma_f^{\frac{1}{2}}$, and since P_k is symmetric,

$$\Sigma_{\text{filter}} = P_k \Sigma_f P_k = P_k \Sigma_f^{\frac{1}{2}} \Sigma_f^{\frac{1}{2}} P_k = \left(\Sigma_f^{\frac{1}{2}} P_k \right)^T \left(\Sigma_f^{\frac{1}{2}} P_k \right) \succcurlyeq 0,$$

so $\mathbf{u}^T \Sigma_{\text{filter}} \mathbf{u} \geq 0$ for every $\mathbf{u} \in \mathbb{R}^F$.

Strict positive-definiteness after perturbation. Evaluate the quadratic form of the full regularised operator for an arbitrary non-zero vector $\mathbf{u} \in \mathbb{R}^F \setminus \{\mathbf{0}\}$:

$$\mathbf{u}^T \Sigma_{f, \text{reg}} \mathbf{u} = \mathbf{u}^T \Sigma_{\text{filter}} \mathbf{u} + \sigma^2 \mathbf{u}^T I_F \mathbf{u} = \mathbf{u}^T \Sigma_{\text{filter}} \mathbf{u} + \sigma^2 \|\mathbf{u}\|^2 \geq \sigma^2 \|\mathbf{u}\|^2 > 0,$$

where the first inequality uses $\mathbf{u}^T \Sigma_{\text{filter}} \mathbf{u} \geq 0$ and the second uses $\sigma^2 > 0$ together with $\|\mathbf{u}\|^2 > 0$. Because the quadratic form is strictly positive for every non-zero $\mathbf{u}$, the operator $\Sigma_{f, \text{reg}}$ is strictly positive-definite.

Equivalently, in the orthonormal eigenbasis of the symmetric matrix Σ_{filter} , let its eigenvalues be $\theta_1 \geq \dots \geq \theta_F \geq 0$ (with at least $F - k$ of them zero). Adding a multiple of the identity leaves the eigenvectors unchanged and shifts every eigenvalue uniformly, so $\Sigma_{f, \text{reg}} = \Sigma_{\text{filter}} + \sigma^2 I_F$ has eigenvalues $\theta_j + \sigma^2 \geq \sigma^2 > 0$. Consequently every eigenvalue of $\Sigma_{f, \text{reg}}$ is bounded below by σ^2 , its determinant

$$\det(\Sigma_{f, \text{reg}}) = \prod_{j=1}^F (\theta_j + \sigma^2) > 0,$$

is strictly positive, and the inverse $\Sigma_{f, \text{reg}}^{-1}$ exists. ■

Interpretation. Truncating the model to k gradients leaves $F - k$ directions with exactly zero projected variance. In a Mahalanobis form these zero-variance directions appear as

vanishing denominators, so any individual displaying even a slight deviation along them would render the distance undefined. The isotropic term $\sigma^2 I_F$ installs a uniform variance floor across all F directions — absorbing residual environmental variation, minor landmark-placement error, and structure not captured by the core manifold — and thereby guarantees a well-conditioned, invertible operator, keeping the model stable even when it assesses highly atypical faces.

6. Facet-Level Idiosyncrasy: the Mahalanobis Score

With a strictly positive-definite, invertible baseline operator in hand, we quantify how far an individual’s connectivity signature for a given facet departs from the cohort. The **facet-level idiosyncrasy score** of individual i at facet f is the Mahalanobis distance of that individual’s profile from the population mean under the regularised covariance:

$$D_M(i, f) = \sqrt{(\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)^T \Sigma_{f, \text{reg}}^{-1} (\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)} \in \mathbb{R}.$$

The score $D_M(i, f)$ is a non-negative scalar. It measures the deviation of the individual’s global wiring diagram for facet f from the Northwestern Han Chinese population manifold, with the inverse covariance weighting the deviation according to the cohort’s correlation structure and thereby accounting for multi-facet co-dependencies. By Lemma 2 the inverse $\Sigma_{f, \text{reg}}^{-1}$ is guaranteed to exist, so the score is always well defined.

7. Aggregation to a Global Individual-Level Index

To distil the entire three-dimensional facial canvas into a single phenotypic index for individual i , we pool the facet-level scores across all $F = 7,025$ facets by root-mean-square (RMS) aggregation:

$$\mathcal{I}_i = \sqrt{\frac{1}{F} \sum_{f=1}^F [D_M(i, f)]^2} \in \mathbb{R}.$$

The global idiosyncrasy score $\mathcal{I}_i$ is a single non-negative scalar summarising the overall structural distinctiveness of individual i 's face. The RMS form is chosen deliberately over a simple average: squaring before summation preserves the contribution of localised, high-intensity clusters of morphological variation — for example, unique structural configurations confined to the midface or mandible — so that such focal signals retain their integrity and are not diluted by the large number of near-baseline facets.

8. Synthesis and Well-Posedness of the Pipeline

Combining the results above yields a fully specified and mathematically stable pipeline. Lemma 1 establishes that the first term of $\Sigma_{f, \text{reg}}$ is an orthogonal projection of the empirical covariance onto the intrinsic, non-linear facial manifold, purifying the baseline of noise directions while preserving authentic anatomical variance. Lemma 2, together with the Residual Variance Floor proposition, establishes that the isotropic term renders the operator strictly positive-definite and globally invertible, with all eigenvalues bounded below by σ^2 .

Consequently the facet-level Mahalanobis score

$$D_M(i, f) = \sqrt{(\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)^T \Sigma_{f, \text{reg}}^{-1} (\mathbf{c}_{i,f} - \boldsymbol{\mu}_f)}$$

is protected against the rank-deficiency instability that would otherwise arise from $F > N$, and the aggregated index $\mathcal{I}_i$ inherits this stability. Both quantities therefore measure structural idiosyncrasy relative to a rigorous, non-linear baseline distribution of the North-western Han Chinese cohort, ensuring that the downstream deep-learning connectivity features reflect authentic morphological divergence rather than geometric or numerical artefact.